

TripWiz — Folder 06: Cost Calculation

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How to read this document

This document contains the complete cost estimation for the TripWiz production architecture. It is structured in four parts:

- 1. **Foundational Assumptions** — every sizing parameter, its rationale, and the source of the estimate.
- 2. **AWS Resource Inventory** — every service in the deployed architecture, confirmed directly from `backend/infra/template.yaml` and `frontend/amplify.yaml`.
- 3. **Input Parameters & Derivation** — the exact numbers fed into the AWS Pricing Calculator for each service, derived from the assumptions.
- 4. **Cost Summary** — monthly and annual totals for year 1 (with AWS free tiers) and steady-state (after free tiers expire), with cost-driver analysis.

Official calculator export: The live AWS Pricing Calculator estimate is published at the following shareable link and was exported as a PDF for submission:

<https://calculator.aws/#/estimate?id=6b4f45da2a61d16955e4a5e11799ed08cc26fb87>

Part 1 — Foundational Assumptions (הנחות יסוד)

Before calculating anything, we established realistic sizing parameters for a B2C travel-planning SaaS in its first year of operation. These figures are intentionally conservative so the estimate represents a credible baseline rather than a best-case scenario.

1.1 User Base

Parameter	Estimate	Rationale
Registered users at launch (month 1)	1,000	Early-adopter + initial marketing campaigns
Registered users at year-end	35,000	Midpoint of 20K–50K growth target
Monthly Active Users (MAU)	7,000	20% of 35K — travel apps are inherently seasonal and sporadic
Peak concurrent users (CCU)	350	5% of MAU — weekend evenings or pre-summer surge
Month-over-month growth rate	17.5%	Midpoint of healthy 15–20% MoM target

1.2 Usage Patterns (per active user per month)

Parameter	Estimate	Rationale
Trips created per active user/month	1.5	Tourists travel 1–3×/year; active users create 1–2 trips
Itinerary stops per trip	17	Typical 3–5 day itinerary: hotels, attractions, restaurants, flights

AI Optimize calls per trip	4	Iterative trial-and-error until the user is satisfied with the route
Weather Validate calls per trip	2.5	Usually triggered near departure or after stop changes
Place searches per planning session	20	Searching for points of interest is the most frequent in-app action
Live co-edit WebSocket sessions per trip	2	Collaborative planning between travel partners
Average WebSocket session duration	22 min	Midpoint of 15–30 min observed range

1.3 Storage (per user / per trip)

Parameter	Estimate	Rationale
File attachments uploaded per trip (S3)	4 files	Flight tickets, hotel vouchers, attraction passes
Average file size	1.75 MB	PDFs and images — midpoint of 1.5–2 MB range
DynamoDB trip item size	20 KB	JSON with 17 stops, metadata, permissions, S3 keys (no binary content)

1.4 Notifications & Scheduling

Parameter	Estimate	Rationale
Trip reminder opt-in rate	75%	High opt-in expected for schedule-tied travel notifications
Weather alert SNS messages per day	500	~5% of 10K active-traveller-day users have active trips on any given day

1.5 Derived Monthly Volumes (end of year 1)

These secondary figures are computed from the assumptions above and fed directly into the pricing calculator.

Derived figure	Calculation	Value
New trips per month	$7,000 \text{ MAU} \times 1.5$	10,500
Bedrock invocations per month	$7,000 \times 1.5 \times 4$	42,000
Place search requests per month	$7,000 \times 1.5 \times 20$	210,000
Route calculations per month	$7,000 \times 1.5 \times 1$	10,500
WebSocket sessions per month	$7,000 \times 1.5 \times 2$	21,000
WebSocket messages per month	$21,000 \text{ sessions} \times 50 \text{ msgs}$	1,050,000
Connection-minutes per month	$21,000 \times 22 \text{ min}$	462,000

S3 new storage per month	$7,000 \times 1.5 \times 4 \times 1.75 \text{ MB}$	73.5 GB
S3 cumulative storage (year-end)	avg ~40 GB/month $\times$ 12	~480 GB
S3 PUT requests per month	$7,000 \times 1.5 \times 4$	42,000
S3 GET requests per month	$42,000 \times 3$ downloads	126,000
Reminder emails per month	$350 \text{ trips/day} \times 2 \text{ days} \times 75\% \times 30$	15,750
Weather-alert admin emails per month	$500/\text{day} \times 30$	15,000
Collaboration-invite emails per month	$7,000 \times 30\% \times 2$	4,200
Total SES emails per month		~35,000
Lambda invocations per month (all functions)	see Part 3	~1,886,000
Lambda GB-seconds per month	see Part 3	~245,000

Part 2 — AWS Resource Inventory

Every resource below is declared in `backend/infra/template.yaml` (SAM) or `frontend/amplify.yml`. Nothing is assumed or inferred.

#	AWS Service	Resource name in template	Purpose
1	AWS Lambda	TripWiz-Trips	Main REST API handler (all trip, place, admin, and user-pref routes)
2	AWS Lambda	TripWiz-Validate	On-demand + nightly weather validation
3	AWS Lambda	TripWiz-WebSocketHandler	Real-time co-editing WebSocket router
4	AWS Lambda	TripWiz-WebSocketAuthorizer	JWT validator for WebSocket <code>\$connect</code>
5	AWS Lambda	TripWiz-TripReminder	Daily scheduled trip-reminder email sender
6	AWS Lambda	TripWiz-SnsToSes	SNS weather-alert → SES email bridge
7	AWS Lambda	TripWiz-PostConfirmation	Cognito post-confirmation trigger — seeds default notification prefs for new sign-ups
8	API Gateway (REST)	TripWizApi	REST API — 29 routes, Cognito-authorized
9	API Gateway (WebSocket)	TripWizWebSocketApi	WebSocket API — <code>\$connect</code> , <code>\$disconnect</code> , <code>join</code> , <code>leave</code> , <code>edit</code> , <code>cursor</code> , <code>ping</code>
10	Amazon DynamoDB	TripWizTable	Single-table design, PAY_PER_REQUEST billing, TTL enabled, GS11 for date queries

11	Amazon Cognito	TripWizUserPool + TripWizWebClient + Admins group	User registration, SRP auth, admin RBAC
12	Amazon Bedrock	Foundation model us.anthropic.claude-haiku-4-5-20251001-v1:0	AI-powered itinerary route optimisation
13	Amazon Location Service	TripWizPlaceIndex (Esri, SingleUse)	Geocoding and place-search
14	Amazon Location Service	TripWizRouteCalculator (Esri)	Road-following route calculation
15	Amazon S3	TripWizDocumentsBucket	Trip attachment storage (presigned PUT/GET URLs)
16	Amazon SES	— (via SDK)	Transactional emails: reminders + weather alerts
17	Amazon SNS	TripWiz-Alerts	Weather-alert fan-out from ValidateLambda → SnsToSes
18	Amazon EventBridge	NightlyTrigger (rate 1 day) + DailyReminder (cron 0 8 * * ? *)	Scheduled Lambda invocations
19	AWS Amplify	frontend/amplify.yml	React/Vite frontend hosting + CDN + CI/CD pipeline
20	AWS Secrets Manager	TripWiz/MappingApiKey + TripWiz/OpenWeatherApiKey	Secure API key storage, retrieved by Lambdas at cold start
21	Amazon CloudWatch Logs	Auto-created per Lambda	Lambda execution logs

Part 3 — Input Parameters & Derivation per Service

Lambda (7 functions)

Function	Invocations/month	Avg duration	Memory	GB-seconds
TripWiz-Trips	700,000	500 ms	512 MB	179,200
TripWiz-WebSocketHandler	1,050,000	100 ms	128 MB	13,440
TripWiz-WebSocketAuthorizer	42,000	200 ms	128 MB	1,075
TripWiz-Validate	79,000	2,500 ms	256 MB	50,400
TripWiz-TripReminder	30	30,000 ms	256 MB	230
TripWiz-SnsToSes	15,000	200 ms	128 MB	384
TripWiz-PostConfirmation	500	150 ms	128 MB	9

Total	1,886,530			244,738
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TripWiz-PostConfirmation derivation (500/mo): ~500 new sign-up confirmations per month against the 7,000 MAU base; single conditional `PutItem` to DynamoDB, ~150 ms.

REST invocations derivation (700K): 7,000 MAU × ~100 API calls/user/month (list trips ×30, stop operations ×25, place searches ×30, AI/validate/misc ×15).

Validate invocations (79K): 26,250 on-demand (7K × 1.5 × 2.5) + 52,500 scheduled (1,750 upcoming trips/day × 30 days).

Permanent free tier: 1M requests/month + 400,000 GB-s/month. 244,729 GB-s < 400,000 → **Lambda GB-second cost = \$0**. Only 886K requests over free tier are billable.

API Gateway

The AWS Pricing Calculator combines REST and WebSocket into a single line item.

Input	Value
REST API requests/month	700,000
Cache memory size	None
Average REST request size	34 KB
WebSocket messages/month	1,050,000
Average WebSocket message size	32 KB
Total connection-minutes/month	472,500

DynamoDB (On-Demand / PAY_PER_REQUEST)

Input	Derivation	Value
Average item size (all attributes)	Conservative calculator input (actual trip items ~20 KB)	1 KB
Storage (GB)	126K trips + users + access items	3 GB

Permanent free tier: 25 GB storage → storage cost = \$0. Billing mode declared in template: `PAY_PER_REQUEST`.

Using 1 KB average item size in the calculator is a conservative simplification; the calculator derives request-unit costs from storage size and item size internally.

Amazon Cognito

Input	Value
MAU	7,000
Free tier threshold	50,000 MAU

Cost = \$0 (well under the 50K MAU permanent free tier).

Amazon Bedrock — Claude Haiku 4.5

Model ID (from template): `us.anthropic.claude-haiku-4-5-20251001-v1:0`

Input	Derivation	Value
Inference route	AWS Pricing Calculator selection	Global
Inference type	AWS Pricing Calculator selection	On-Demand — Standard
Avg requests per minute	42,000 calls/month ÷ (30 days × 23 active hrs × 60 min)	1 req/min
Active hours per day	Excluding quiet hours (1:00–2:00 AM)	23 hrs/day
Avg input tokens/call	System prompt + 17-stop itinerary JSON	1,200 tokens
Avg output tokens/call	Optimised route with explanations	600 tokens
Total input tokens/month	1 req/min × 60 × 23 × 30 × 1,200	~49.7M tokens
Total output tokens/month	1 req/min × 60 × 23 × 30 × 600	~24.8M tokens

Amazon Location Service (Esri)

Input	Derivation	Value
Place Index search requests/month	7K × 1.5 × 20	210,000
Route Calculator requests/month	7K × 1.5 × 1	10,500
Data source	Declared in template	Esri
Intended use	Declared in template	SingleUse

12-month free tier: 3M geocoding + 3M routing requests/month → Year 1 cost = \$0.

Amazon S3

Input	Value
Standard storage (year-end cumulative)	480 GB
PUT requests/month	42,000
GET requests/month	126,000
Data transfer	Negligible (presigned URLs served directly to client)

Amazon SES

Input	Value
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Emails sent/month	35,000
Email types	HTML (reminders) + plain text (alerts)

Free tier: 3,000 emails/month → 32,000 billable.

Amazon SNS

Input	Value
Publish requests/month	19,200
Delivery target	AWS Lambda

Lambda delivery is free under SNS pricing → cost = \$0.

Amazon EventBridge

Input	Value
Scheduled rules	2
Invocations/month	60

Free tier: 14M scheduler invocations/month → cost = \$0.

AWS Amplify (Frontend Hosting + CDN + CI/CD)

Confirmed by `frontend/amplify.yml`. The SAM template has no standalone S3 static hosting bucket or separate CloudFront distribution for the frontend — Amplify provides both as a managed service.

Input	Value
Build minutes/month	20 (10 deploys × ~2 min/build)
Hosting storage	~15 MB (Vite <code>dist/</code> bundle)
Data transfer out/month	35 GB (7,000 MAU × 5 MB/session)

12-month free tier: 1,000 build-min + 5 GB storage + 15 GB transfer/month.

AWS Secrets Manager

Input	Value
Secrets	2 (TripWiz/MappingApiKey, TripWiz/OpenWeatherApiKey)
API calls/month	~200 (Lambda cold-start retrievals; value cached per execution environment)

Amazon CloudWatch Logs

Input	Value
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Log data ingested/month	~0.95 GB (1.886M invocations × ~500 B/log)
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Permanent free tier: 5 GB ingestion/month → cost = \$0.

Part 4 — Cost Summary

All figures below are taken directly from the **official AWS Pricing Calculator export** (PDF, exported 06/05/2026). The calculator applies permanent free tiers (Lambda, Cognito, DynamoDB storage, SNS, EventBridge, CloudWatch) but does **not** apply 12-month free tiers (REST API Gateway, Amazon Location Service, Amplify data transfer). A separate year-1 column shows those savings.

Monthly Cost Breakdown

Service	Official calculator/month	Year 1/month	Notes
AWS Lambda (all 7 functions)	\$0.01	\$0.01	Permanent free tier covers all GB-s; ws_handler has \$0.01 overage on requests
Amazon API Gateway (REST + WebSocket)	\$3.50	\$1.17	Year 1: REST 700K calls within 1M free-tier; only WebSocket billed
Amazon DynamoDB	\$4.75	\$4.75	Permanent free tier covers 25 GB storage; request units billed
Amazon Cognito	\$0.00	\$0.00	7K MAU permanently free (threshold: 50K MAU)
Amazon Bedrock	\$173.88	\$173.88	Dominant cost — Global inference, On-Demand Standard
Amazon Location Service	\$110.25	\$0.00	Year 1: 12-month free tier (3M requests/month) covers all 210K searches
Amazon S3	\$11.30	\$11.30	480 GB storage + 42K PUT + 126K GET
Amazon SES	\$3.20	\$3.20	32K billable emails (3K/month permanently free)
Amazon SNS	\$0.00	\$0.00	Lambda-target deliveries are free
Amazon EventBridge	\$0.00	\$0.00	60 invocations vs 14M free
AWS Amplify	\$5.45	\$3.20	Year 1: 15 GB/month transfer free → only 20 GB billed
AWS Secrets Manager	\$0.80	\$0.80	2 secrets × \$0.40/month
Amazon CloudWatch	\$0.00	\$0.00	0.95 GB ingestion vs 5 GB/month free
Monthly Total	\$313.14	~\$198.31	
Annual Total	\$3,757.68	~\$2,380	

The official \$3,757.68 annual figure is sourced directly from the PDF export and represents $12 \times \$313.14$. The year-1 figure deducts 12-month free-tier savings.

Cost-Driver Analysis

Amazon Bedrock (AI route optimisation):	\$173.88/month	→ 55.5% of calculator total
		→ 87.7% of year-1 cost
Amazon Location Service:	\$110.25/month	→ 35.2% of calculator total
	\$ 0.00/month	→ year 1 (12-month free tier)
Everything else combined:	\$ 28.01 /month	→ 8.9% of calculator total
	\$ 24.43 /month	→ 12.3% of year-1 cost

Key observations:

- 1. Bedrock is the dominant and constant cost.** At \$173.88/month it accounts for over half the total bill in all scenarios and is unaffected by any free tier. It scales linearly with AI optimize usage. Reducing the per-trip call limit from 4 to 3 saves approximately \$43/month (~25%).
- 2. Amazon Location Service creates a sharp year-2 pricing cliff.** The 12-month free tier covers our 210K monthly searches with 93% headroom to spare, making year 1 essentially free for this service. After expiry it jumps to \$110.25/month — the second-largest line item. A client-side geocoding cache deduplicating repeated searches within a session can reduce billable requests by an estimated 30–40%, lowering this to ~\$65–\$80/month.
- 3. The serverless architecture delivers on its cost promise.** Lambda costs \$0.01, DynamoDB \$4.75, API Gateway \$3.50, and CloudWatch \$0.00 — all six backend compute services combined are under \$10/month. The platform handles demand spikes without provisioned capacity changes.
- 4. Cognito is permanently free at this scale.** At 7,000 MAU we are 14% of the 50,000 MAU free threshold. It only becomes a line item if TripWiz grows beyond ~55,000 monthly active users.

Monthly Cost Projection (Year 1 growth curve)

Bedrock cost scales proportionally with MAU ($\$173.88 \div 7,000 \text{ MAU} = \sim\$24.84 \text{ per } 1,000 \text{ MAU/month}$). All other services are within free tiers or near-fixed through most of year 1.

Month	Registered Users	MAU	Bedrock	Other services	Est. Monthly Cost
1	1,000	200	~\$5	~\$10	~\$15
3	1,372	470	~\$12	~\$12	~\$24
6	2,436	1,000	~\$25	~\$15	~\$40
9	4,321	3,100	~\$77	~\$20	~\$97
12	7,662	7,000	~\$174	~\$24	~\$198

The ramp is gentle in months 1–6 because Bedrock dominates and MAU is still low. The 12-month free-tier savings on Location Service and REST API Gateway keep non-AI costs near-zero for the entire first year.